

Photon condensation and macroscopic quantum tunnelling of the persistent current in a π loop

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A superconducting loop closed by a π junction carries a spontaneous persistent current when its inductance exceeds the Josephson inductance of the junction. We ask what happens when that inductance belongs to a quantum LC mode and the π junction is an interacting quantum dot, so that the loop is a circuit-QED element and the flux a quantum variable. Solving the full electron–photon problem by DMRG, we show that the current appears as a static displacement of the mode, i.e. a photon condensate, when the dot is in its doublet (π) state and the screening parameter $\beta = 2\pi LI_c/\Phi_0$ exceeds one. The criterion is the single-loop form of the photon-condensation instability, which the flux coupling is allowed to reach because the π shift makes the orbital response paramagnetic. Quantum fluctuations of the flux do not drive the transition but oppose it: the exact ground state is a cat of the two current directions, split by a tunnelling gap $\delta \propto e^{-E_J/\hbar\omega}$. We map the phase diagram in the interaction and coupling strengths, show where the self-consistent product-state (mean-field) description holds and where it fails, and give the scales for quantum-dot, kinetic-inductance and SFS realizations.

Introduction. A superconducting ring interrupted by a π junction has a ground state that carries a current without any applied magnetic field [1]. The junction prefers a Cooper-pair phase difference of π , the loop can only supply it through a flux $\Phi_0/2$, and when the loop inductance L is large enough that the magnetic cost is below the Josephson gain, $\beta \equiv 2\pi LI_c/\Phi_0 > 1$, the ring settles into one of two degenerate states carrying opposite currents $\pm I$. Such half-flux states are textbook evidence of d -wave pairing [2] and of the π shift in superconductor–ferromagnet–superconductor (SFS) junctions [3, 4]; in all of them the flux is a classical variable, because E_J exceeds every other energy scale by orders of magnitude.

Two developments make it natural to revisit this problem as a quantum one. First, π junctions can now be made from gate-tunable quantum dots, where the π shift is produced by a single unpaired spin and the Josephson energy is a few GHz [5–9]. Second, the loop inductance can be the inductance of a superconducting LC resonator, so that the loop is a circuit-QED element [10] whose flux is the quadrature of a photon mode with $\hbar\omega \sim E_J$. The ring current then becomes a static displacement of the mode. Whether a cavity field can condense in the ground state of a light–matter system is a debated question: a no-go theorem forbids it for a uniform field coupled to the charge [11, 12], while a field coupling to the orbital magnetization can condense if the matter has a paramagnetic orbital response [13]. A π loop is the simplest system in which that response is paramagnetic by construction.

In this Letter we solve the problem without approximation. We treat a lattice model of a superconducting loop closed by an Anderson impurity and threaded by the flux of a quantum LC mode by DMRG on the full electron–photon Hilbert space, and compare with the adiabatic (Born–Oppenheimer) reduction and with the self-consistent product-state mean field that is customarily used for such problems. We find (i) a phase

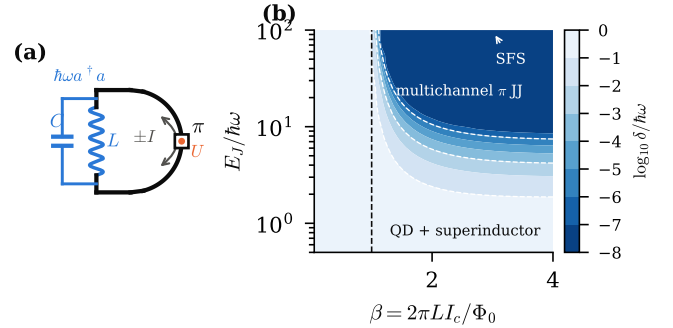


FIG. 1. (a) A superconducting loop closed by an interacting quantum dot (U) shares its inductance L with an LC mode of frequency ω . For $U > U_c$ the dot is a π junction and the loop has two degenerate current-carrying states $\pm I$. (b) Tunnel splitting δ between these two states, from the adiabatic Hamiltonian (3) with the DMRG $E(\varphi)$ of the $U = 2t$ junction, in the plane of the screening parameter $\beta = 2\pi LI_c/\Phi_0$ and the quantum parameter $E_J/\hbar\omega$. White lines: $\delta/\hbar\omega = 10^{-1}, 10^{-3}, 10^{-6}$; dashed: classical threshold $\beta = 1$. Labels: regimes of the platforms discussed at the end.

diagram in the plane of the dot interaction U and the coupling g whose two boundaries are the $0-\pi$ transition of the dot and the classical screening criterion $\beta = 1$; (ii) a photon condensate of $\langle a^\dagger a \rangle \simeq 8$ carrying a current of $0.95 I_c$, whose direction is chosen spontaneously because the Hamiltonian is time-reversal symmetric; (iii) a ground state that is, strictly, a cat of the two current directions, whose tunnel splitting δ vanishes as $e^{-E_J/\hbar\omega}$ and sets the only energy scale between them; and (iv) the condition $E_J \gg \hbar\omega$ under which mean field is reliable, together with an explicit example where it predicts a current that the exact solution does not have.

Model. We consider the circuit of Fig. 1(a): a superconducting (SC) chain of $L - 1$ sites closed into a loop by a quantum dot (site 0), with the loop threaded by the

flux of a single lumped LC mode,

$$\begin{aligned}
H = & \hbar\omega a^\dagger a - t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c_{r+1\sigma}^\dagger c_{r\sigma} + \text{h.c.} \right) + \Delta \sum_{r=1}^{L-1} \left(c_{r\uparrow}^\dagger c_{r\downarrow}^\dagger + \text{h.c.} \right) \\
& + \epsilon_d n_0 + U n_{0\uparrow} n_{0\downarrow} - t_J \sum_{\sigma} \left(c_{0\sigma}^\dagger c_{L-1\sigma} + \text{h.c.} \right) \\
& - t_J \sum_{\sigma} \left(e^{i\varphi_{\text{cl}}} e^{ig(a+a^\dagger)} c_{1\sigma}^\dagger c_{0\sigma} + \text{h.c.} \right). \quad (1)
\end{aligned}$$

The first line is the mode and a BCS chain with hopping t and a fixed pair field Δ . The second line is the Anderson impurity (level $\epsilon_d = -U/2$, on-site repulsion U) coupled by t_J to the two ends of the chain. The third line carries the total Peierls phase of the loop. Three remarks fix its meaning. (i) The phase is a loop property and can be placed on a single bond by a gauge choice; Eq. (1) is identical to a uniform vector potential along the ring with the pair field transformed covariantly, $\Delta_r \rightarrow \Delta e^{-2i\theta_r}$ [14, 15], whereas a uniform A with a real Δ is a different model [16]. (ii) The mode enters through the displacement operator $e^{ig(a+a^\dagger)}$, so diamagnetic (A^2) and all higher terms are included. (iii) $g = 2\pi\Phi_{\text{zpf}}/(h/e)$ is the *electron* phase produced by the zero-point flux of the mode; one SC flux quantum $\Phi_0 = h/2e$ corresponds to an electron phase π . For a lumped resonator of impedance Z in series with the loop, $(2g)^2 = \pi Z/R_Q$, $R_Q = h/4e^2$. A fixed, real Δ means that the condensate phase of the arms is pinned (a proximitized ring); the kinetic inductance of a self-supporting ring then simply adds to L , see below. Unless stated otherwise $t = 1$, $\Delta = t$, $t_J = 0.5t$, $L = 14$, $\hbar\omega = 7.6 \times 10^{-4}t$ and there is *no* static flux: $\varphi_{\text{cl}} \rightarrow 0^+$, with a seed $\varphi_{\text{cl}} = 10^{-3}$ (or 10^{-5}) used only to select one of two degenerate states. We write Eq. (1) compactly as $H = \hbar\omega a^\dagger a + H_{\text{ring}}[\varphi_{\text{cl}} + g(a+a^\dagger)]$.

The only property of the ring that will matter is its ground-state energy at fixed phase,

$$E(\varphi) = \sum_{m \geq 0} E_m \cos m\varphi, \quad (2)$$

where even harmonics are Cooper-pair ($h/2e$) processes and E_1 is the single-electron (h/e) persistent current of a mesoscopic ring [17], suppressed as $e^{-L/\xi}$. A 0-junction has $E_2 < 0$, a π -junction $E_2 > 0$. Since $H_{\text{ring}}[0]$ is real, $E(\varphi) = E(-\varphi)$ exactly: the two current directions are related by time reversal. This is what distinguishes an intrinsic π junction from a 0-junction biased by $\Phi_0/2$: the latter maps $E_2 \rightarrow -E_2$ but turns the odd harmonics into an explicit tilt $-E_1 \sin \varphi$ that *selects* the current direction (Supplemental Material, SM [16]).

Adiabatic reduction and the classical criterion. When $\hbar\omega$ is well below the Andreev gap E_A of the ring, the electrons follow the flux adiabatically and Eq. (1) reduces, to leading order in $\hbar\omega/E_A$, to a single-particle problem

for the dimensionless flux $X = a + a^\dagger$,

$$H_{\text{BO}} = \frac{\hbar\omega}{4} (P^2 + X^2) + E(\varphi_{\text{cl}} + gX), \quad [X, P] = 2i. \quad (3)$$

This is the Hamiltonian of an rf-SQUID [18] whose inductive energy is supplied by the mode: in the Cooper-pair phase $\theta = 2gX$ it reads $4E_C n^2 + \frac{1}{2}E_L \theta^2 + E(\theta/2)$ with $E_L = \hbar\omega/8g^2$ and $E_C = \hbar\omega g^2$, the fluxonium form [19]. The classical energy $\mathcal{F}(X) = \hbar\omega X^2/4 + E(gX)$ develops two minima $\pm X^*$ when the orbital response of the ring at $\varphi = 0$ is paramagnetic and exceeds the stiffness of the mode,

$$\beta \equiv \frac{2g^2|E''(0)|}{\hbar\omega} > 1, \quad E''(0) < 0. \quad (4)$$

For a pure π junction $E''(0) = -4E_J$, and $\beta = 8g^2 E_J/\hbar\omega = 2\pi L I_c/\Phi_0$ with $I_c = 2eE_J/\hbar$ is the screening parameter of Ref. [1]. Equation (4) is the single-mode form of the photon-condensation criterion of Ref. [13]: a static field $\langle a \rangle \neq 0$ appears because the mode couples to the flux (the curl of $\mathbf{A}$), to which the no-go theorem for a uniform field does not apply, and because the π shift makes the response paramagnetic. A 0-junction, with $E''(0) > 0$, never orders.

The static field is a persistent current. Indeed $\langle [H, P] \rangle = 0$ in any eigenstate gives the exact identity

$$\langle I \rangle = \frac{\hbar\omega}{2g} \langle a + a^\dagger \rangle, \quad I \equiv -\frac{e}{\hbar} \partial_\varphi H_{\text{ring}}, \quad (5)$$

which says that the current around the loop *is* the displacement of the mode, and a photon number $\langle a^\dagger a \rangle \simeq (X^*/2)^2$ is the condensate. We use Eq. (5) to extract the current from DMRG, where the one-site observable $\langle X \rangle$ converges far better than the three-site bond current.

Two remarks complete the classical picture. First, Eq. (4) contains the total loop inductance. If the mode is coupled to the loop by a mutual inductance M rather than galvanically, minimizing the static energy over the resonator flux gives a loop energy $\Phi_r^2/2L_r$ with $L_r \geq M^2/L_c$, so a resonator cannot lower the static threshold; Eq. (1) is the galvanic limit. Likewise, if the arms of the ring are a self-supporting superconductor their kinetic inductance L_K adds to L : for our 14-site chain with Δ determined self-consistently, we find $\beta_K = 2\pi L_K I_c/\Phi_0 \simeq 0.4$ at $U = 2t$ [16], so the mode has to supply only the remainder. Second, and central to what follows, the zero-point fluctuations of the mode do *not* drive the transition; they oppose it. At finite $E_J/\hbar\omega$ the two minima are connected by tunnelling, the ground state is the symmetric cat $(|\odot\rangle + |\oslash\rangle)/\sqrt{2}$, $\langle I \rangle = 0$, and the two current states are split by $\delta \sim \hbar\omega e^{-S}$ with an instanton action $S \propto (E_J/\hbar\omega) s(\beta)$. A sharp transition with mean-field exponents is recovered only in the limit $E_J/\hbar\omega \rightarrow \infty$ at fixed β , where $E_C/E_J = \beta(\hbar\omega/E_J)^2/8 \rightarrow 0$; this is the classical-oscillator limit of the quantum Rabi model [20]. The phase diagram of this crossover is Fig. 1(b).

Methods. We solve Eq. (1) by DMRG (TeNPy [21]) on the full electron–photon space: the ring is stored in folded order next to a bosonic site (Fock cutoff $N_{\text{ph}} = 30\text{--}50$), fermion parity and S^z are conserved, and the vertex $e^{ig(a+a^\dagger)}c_{1\sigma}^\dagger c_{0\sigma}$ is a three-site term. DMRG reproduces exact BdG energies at $g = 0$ to 10^{-14} and, for a 4-site ring with a photon, exact diagonalization and Eq. (5) to 10^{-7} . In production ($\chi = 256\text{--}400$) ground-state energies are converged to $3 \times 10^{-6}t$ and excited states (DMRG orthogonal to the ground state) to $3 \times 10^{-5}t \approx 0.04\hbar\omega$. The (U, g) phase diagram is obtained in two steps that are each exact: DMRG of the ring at fixed phase in both parity sectors gives $E(\varphi; U)$, Eq. (2), on a grid of U (16 values, 9–17 phases each, interpolated in between); then the photon is treated exactly (full Fock space, no product ansatz) in Eq. (3). The only approximation is adiabaticity, $\hbar\omega/E_A \sim 10^{-3}$, which we check against full DMRG at $U = 2t$ and $U = 6t$ below. The mean-field (MF) reference is the self-consistent product state $|\psi_{\text{el}}\rangle \otimes |\psi_{\text{ph}}\rangle$ [22], with Hartree–Fock BdG electrons and the exact photon state in the field of $\langle c_{1\sigma}^\dagger c_0 \rangle$, swept upward in g with a warm start.

The junction: interaction-driven π shift. Figure 2 characterizes the ring without the mode. DMRG in the even and odd fermion-parity sectors reproduces the known singlet–doublet transition of a quantum dot between superconductors [5, 6, 23, 24]: for $U < U_c \simeq 0.9t$ the singlet is the ground state at $\varphi = 0$ and $E_2 < 0$ (a 0-junction), for $U > U_c$ the Kramers doublet is the ground state at all phases and $E(\varphi)$ has its minimum at the Cooper-pair phase π . At $U = 2t$ we find $E_2 = +7.39 \times 10^{-3}t$ with $|E_1|, |E_4| < 10^{-4}t$: a pure π junction with $E_J = |E''(0)|/4 = 7.6 \times 10^{-3}t$, i.e. $E_J/\hbar\omega = 10$ for our mode, and an excitation gap $E_A \simeq 0.26t \gg \hbar\omega$. The π coupling decreases monotonically with U ($E_2 \propto U^{-1.3}$ for $4t \leq U \leq 10t$), as expected for cotunnelling through a singly occupied level; at fixed $\hbar\omega$ the axis U is therefore also an axis of decreasing $E_J/\hbar\omega$, from 15 at $U = t$ to 1.5 at $U = 10t$. The ring is longer than the coherence length ($L \gg \xi \simeq 2t/\Delta$), so the mesoscopic h/e harmonic is negligible; this is essential, because in a short ring E_1 dominates $E(\varphi)$ [Fig. 2(a)] and the instability would be that of the normal persistent current rather than of the Josephson junction [16].

Phase diagram. Figure 3 is the central result. Panel (a) follows the ground state at $U = 2t$ as the coupling grows. For $g < g_c \simeq 0.12$ the mode stays near its vacuum and the current is the linear response to the seed, $|\langle I \rangle| \lesssim 10^{-4}et/\hbar$, scaling with it. At g_c the photon number jumps to $\langle a^\dagger a \rangle \simeq 8$ and a current $|\langle I \rangle| = 1.4 \times 10^{-2}et/\hbar = 0.95 I_c$ appears; reducing the seed by two orders of magnitude leaves it unchanged. The current is spontaneous. Full DMRG of the electron–photon problem (circles) and the exact photon in Eq. (3) (lines) agree to a few percent in g_c , current and photon number, so adiabaticity holds and the two-step procedure

is justified. Beyond its maximum the current decreases as $I \simeq \hbar\omega\varphi^*/2g^2$: the condensate flux saturates at $\Phi_0/2$ (where the junction current vanishes) while the effective inductance $\propto g^2/\hbar\omega$ keeps growing. The classical threshold is $\beta = 1$ at $g = 0.112$; quantum fluctuations shift the observed onset to 0.12–0.13 and round it.

Panel (b) shows $|\langle I \rangle|$ in the (U, g) plane. The ordered region has two boundaries. The lower one, $U_c \simeq 0.9t$, is the $0\text{--}\pi$ transition of the junction itself: below it the ring is a singlet with $E''(0) > 0$, the orbital response is diamagnetic and no current appears at any g [the boundary bends down slightly at large g , where the condensation energy of the doublet sector tips the parity balance]. The left boundary, $g_c(U)$, follows the classical line $\beta(U, g) = 1$ (white dashed), shifted to larger g by the zero-point energy, and moves to stronger coupling as $E_J(U)$ decreases. At $U = t, 2t, 4t$ we find $g_c = 0.10, 0.13, 0.20$ against the classical 0.092, 0.112, 0.155.

The upper boundary of the coloured region is *not* a phase boundary but a crossover controlled by the tunnel splitting, panel (c). Along the U axis $E_J/\hbar\omega$ decreases, and δ , which is $\sim 10^{-8}\hbar\omega$ at $U = 2t$ deep in the ordered phase, grows to $10^{-4}\hbar\omega$ at $U = 4t$, $10^{-2}\hbar\omega$ at $U = 6t$ and $10^{-1}\hbar\omega$ at $U = 8t$. A seed flux φ_{cl} polarizes the cat only if $I^*\varphi_{\text{cl}} \gtrsim \delta$; this is why the two contours in (b) separate for $U \gtrsim 4t$ and why the current of the 10^{-3} seed, $0.95 I_c$ at $U = 2t$, decays to $0.5 I_c$ at $U = 6t$ and $0.15 I_c$ at $U = 7t$. The condensate itself is still there: at $U = 6t$ the lowest doublet carries a flux $\Phi^* = 0.6\Phi_0/2$ and $\langle a^\dagger a \rangle = 2.4$ photons, but the two directions are coherently mixed. Any experiment resolves this crossover with its own scale: a flux offset $\delta\Phi$ or a temperature T selects a current direction when $I^*\delta\Phi$ or $k_B T$ exceeds δ .

Cat state and macroscopic quantum tunnelling. Figure 4 resolves the ordered phase without any seed. Since H is real, $\langle I \rangle = 0$ in every eigenstate and the order is diagnosed by the lowest doublet: the splitting $\delta = E_1 - E_0$ collapses, the flux projected onto the doublet acquires eigenvalues $\pm\Phi^*$, and the flux distribution of the DMRG ground state, extracted from the photon reduced density matrix, turns from a Gaussian into two peaks at $\pm\Phi^*$ whose separation follows the classical minimum [panel (c)]. At $E_J/\hbar\omega = 10$ DMRG and Eq. (3) agree on Φ^* to 3% for all β and on $\delta/\hbar\omega = 0.71$ and 0.28 at $\beta = 0.5$ and 1; deeper in the ordered phase δ falls below the DMRG resolution, and the adiabatic theory gives 7×10^{-4} , 2×10^{-6} and 2×10^{-8} at $\beta = 1.5, 2$ and 3. At $U = 6t$ ($E_J/\hbar\omega = 3.2$) the splitting stays resolvable and full DMRG again agrees with Eq. (3) [squares]. With increasing $E_J/\hbar\omega$ the crossover sharpens into a pitchfork at $\beta = 1$ and δ vanishes exponentially [panel (b)]. The residual δ is the macroscopic quantum tunnelling of a persistent current of ~ 10 photons between two time-reversed states; because the two states are related by time reversal, no other energy scale separates them. Each current state is also Kramers degenerate in the spin of the

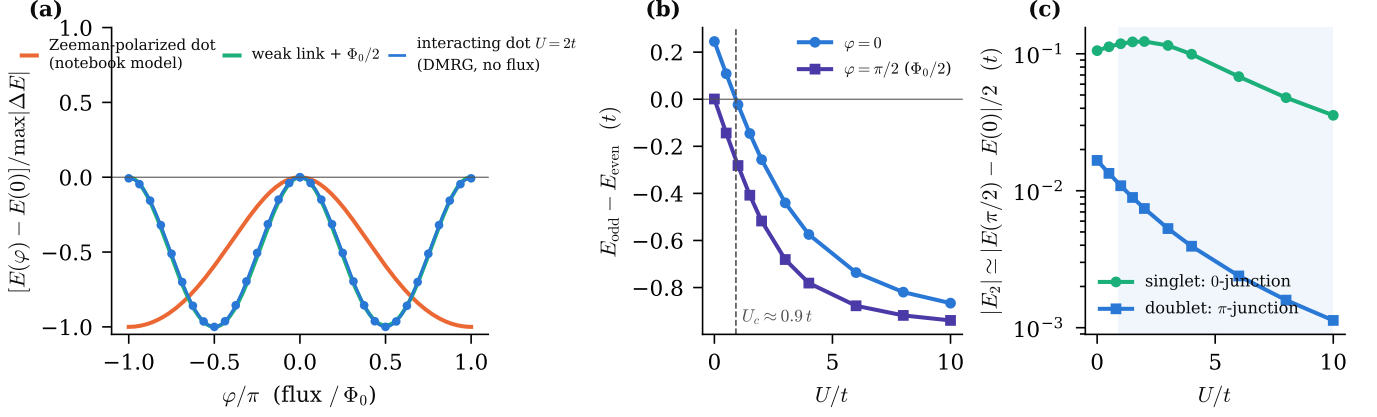


FIG. 2. The junction. (a) Normalized ground-state energy of the loop versus the electron phase φ for three junctions: a Zeeman-polarized dot in a short ring ($L \lesssim \xi$, dominated by the h/e harmonic), an ordinary weak link biased by $\Phi_0/2$, and the interacting-dot ring at $U = 2t$ without any bias (DMRG, symbols; line: cosine fit). (b) Parity competition $E_{\text{odd}} - E_{\text{even}}$ of the singlet (0) and doublet (π) sectors; the doublet is the ground state for $U > U_c \approx 0.9t$. (c) Josephson energy $|E_2|$ in the singlet (0) and doublet (π) sectors; shading: doublet ground state.

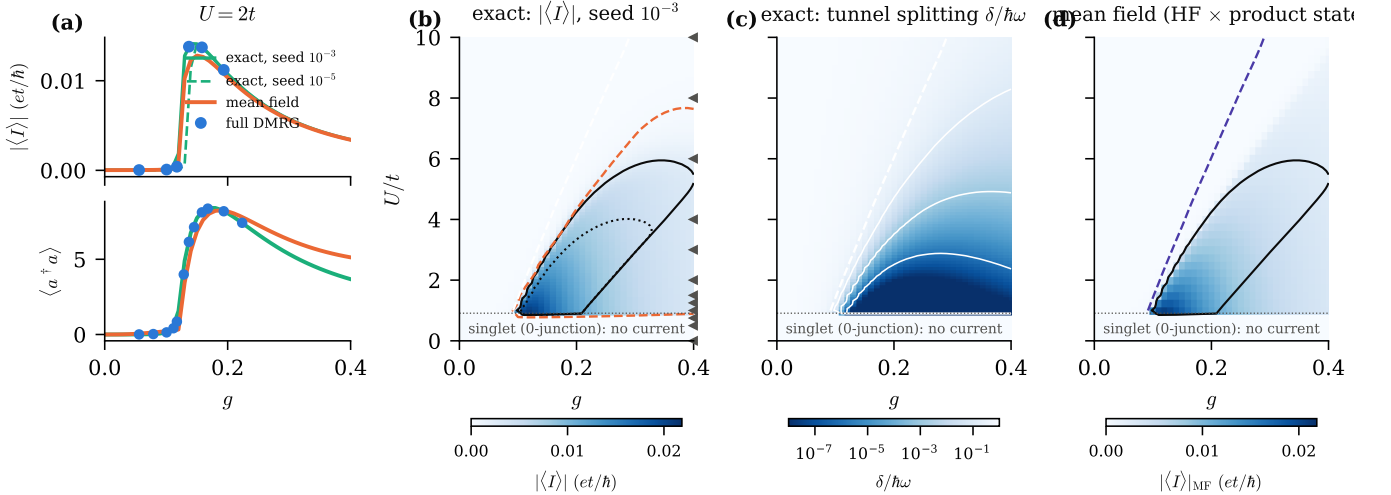


FIG. 3. Phase diagram of Eq. (1) ($L = 14$, $t = 1$, $\Delta = t$, $t_J = 0.5t$, $\epsilon_d = -U/2$, $\hbar\omega = 7.6 \times 10^{-4}t$, no static flux). (a) Current and photon number versus g at $U = 2t$: full DMRG (circles), exact photon in the adiabatic Hamiltonian (3) with the DMRG $E(\varphi)$ (green; solid and dashed: seeds $\varphi_{\text{cl}} = 10^{-3}$ and 10^{-5}), mean field (orange). (b) $|\langle I \rangle|$ in the (U, g) plane for a seed of 10^{-3} ; black contours: $|\langle I \rangle| = I_c/2$ for seeds 10^{-3} (solid) and 10^{-5} (dotted); orange dashed: the same threshold in mean field; white dashed: the classical line $\beta(U, g) = 1$; dotted horizontal: U_c ; triangles: U values computed by DMRG. (c) Tunnel splitting $\delta/\hbar\omega$ of the two lowest states (no seed), same plane; white contours $10^{-1}, 10^{-3}, 10^{-6}$. (d) Mean field, same colour scale as (b), with the exact threshold of (b) overlaid.

dot, so the low-energy manifold is the fourfold multiplet $\{\uparrow, \downarrow\} \otimes \{\circ, \circ\}$ split only by δ .

When does mean field work? Panel (d) of Fig. 3 shows the same map in the product-state mean field. For $U \lesssim 5t$, i.e. $E_J \gtrsim 4\hbar\omega$, it reproduces U_c and $g_c(U)$ to within the grid ($\delta g = 0.01$) and underestimates the maximal current by 10–20%: the product state renormalizes the junction bond by $|\langle e^{ig(a+a^\dagger)} \rangle| < 1$ and thereby loses phase-independent kinetic energy that the exact state recovers by following the flux adiabatically [14, 15]. For $U \gtrsim 6t$ mean field predicts an ordered phase up to $U \simeq 8t$

(orange dashed contour) that the exact solution does not have: there $E_J \lesssim 3\hbar\omega$, the double well is shallow, and the product state, which cannot describe a superposition of the two wells, breaks the symmetry by construction (self-trapping) where the exact ground state is a cat with $\delta \sim 10^{-2} - 10^{-1}\hbar\omega$. The same failure, in extreme form, occurs for the parameters of Ref. [22], a short ring ($L \lesssim \xi$) with a Zeeman-polarized dot and $E_J \simeq \hbar\omega$, $g \simeq 1$: mean field predicts a current above $g \simeq 0.75$, whereas the DMRG ground state lies $1.8\hbar\omega$ lower, keeps a gap of $0.33\hbar\omega$ and carries no current at any g [16]. The

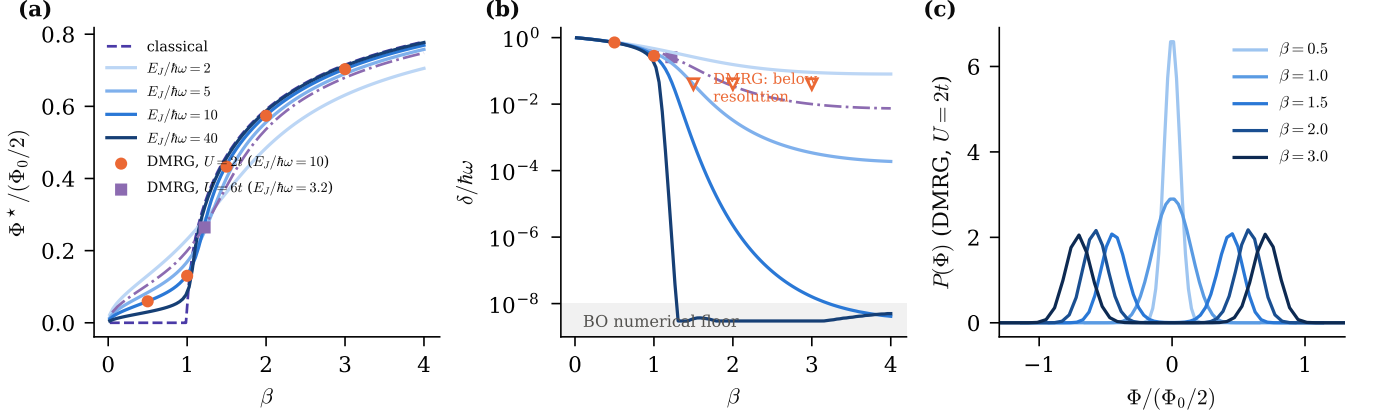


FIG. 4. The cat state ($U = 2t$ unless stated, no flux). (a) Condensate flux Φ^* (eigenvalue of the flux projected onto the two lowest states) versus β for increasing $E_J/\hbar\omega$ (lines: adiabatic theory with the DMRG $E(\varphi)$; dashed: classical); circles: full DMRG at $U = 2t$ ($E_J/\hbar\omega = 10$); squares and dash-dotted line: full DMRG and adiabatic theory at $U = 6t$ ($E_J/\hbar\omega = 3.2$). (b) Tunnel splitting δ between the two current-carrying states. (c) Flux distribution of the DMRG ground state at $U = 2t$, from the photon reduced density matrix: a cat for $\beta > 1$.

product state is reliable when $E_J \gg \hbar\omega$ and $g \ll 1$, i.e. when the mode is nearly classical.

Experimental scales. The two control parameters are $\beta = 2\pi LI_c/\Phi_0$ and $E_J/\hbar\omega$, related by $\beta = 2\pi(Z/R_Q)(E_J/\hbar\omega)$, with $E_J/h \simeq 0.5 \text{ GHz} \times I_c[\text{nA}]$. Three classes of π elements cover Fig. 1(b). (i) SFS junctions ($I_c \sim 1\text{--}100 \mu\text{A}$) reach $\beta > 1$ with geometric inductances of pH to nH and have $E_J/\hbar\omega \sim 10^2\text{--}10^4$: the classical limit, where the half-flux state is fully polarized, as observed [3, 4]. (ii) Quantum-dot π junctions in the doublet regime ($I_c \sim 0.1\text{--}10 \text{ nA}$) need a superinductance $L \sim 0.03\text{--}3 \mu\text{H}$ as in fluxonium [19, 25], the π -qubit geometry of Refs. [26–28]. There $E_J/\hbar\omega \sim 1$, the ground state is the cat with $\delta \sim 0.1\hbar\omega$, and the transition appears as a dip of the $|0\rangle \rightarrow |1\rangle$ frequency and a jump of $\langle 0|\Phi|1\rangle$ in two-tone spectroscopy. (iii) Multichannel or gate-tunable π junctions with $I_c \sim 50\text{--}200 \text{ nA}$ in a high-kinetic-inductance loop ($L \sim 2\text{--}10 \text{ nH}$, $\omega/2\pi \sim 2\text{--}5 \text{ GHz}$) give $\beta \sim 1\text{--}3$ and $E_J/\hbar\omega \sim 10\text{--}30$, the regime of Figs. 3 and 4: a well-defined current of ~ 10 photons whose residual tunnelling $\delta/\hbar\omega \sim 10^{-4}\text{--}10^{-8}$ is a measurable macroscopic-quantum-tunnelling signature. In a 0-junction biased by $\Phi_0/2$ the same state exists, but its direction is selected by the bias, which must then be stable to $|\delta\Phi| \ll \delta/2I^* \lesssim 10^{-6}\Phi_0$; the interaction-driven realization has no such requirement.

Conclusion. A π loop whose inductance is a quantum LC mode is the minimal system in which a cavity field condenses in the ground state. The condensate is the persistent current, its criterion is the classical screening parameter $\beta > 1$ applied to the paramagnetic response of the π junction, and its direction is chosen spontaneously because an interacting dot supplies the π shift without breaking time reversal. The quantum nature of the flux shows up not in the threshold but in the ground

state, a cat of the two currents with a tunnel splitting that vanishes as $e^{-E_J/\hbar\omega}$ and is the only scale between them. The exact solution also delimits the self-consistent product state: it is trustworthy for $E_J \gg \hbar\omega$ and fails, by predicting order that is not there, once the mode is quantum. The same machinery applies to any paramagnetic element in a circuit-QED loop, and the fourfold $\{\sigma\} \otimes \{\odot, \circ\}$ multiplet of the dot realization is a natural starting point for a protected qubit.

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Supplemental Material

S1. GAUGE COVARIANCE AND THE KINETIC INDUCTANCE OF THE ARMS

Gauge. With real Δ on every site, the BdG energy of the ring with the whole Peierls phase φ on the junction bond coincides, to $10^{-10}t$, with that of a uniform vector potential $a_r = \varphi/L$ on all bonds together with the covariant pair field $\Delta_r = \Delta e^{-2i\theta_r}$, $\theta_{r+1} - \theta_r = \varphi\delta_{r0} - a_r$. A uniform a_r with real Δ is not equivalent (at $\varphi = \pi/2$ it differs by $0.13t$): it describes a condensate whose phase does *not* wind, i.e. a different physical situation. Eq. (1) is thus the model of a loop in which the condensate phase of the arms is pinned by a bulk superconductor and the entire gauge-invariant phase drop sits on the junction.

Kinetic inductance. If the chain is instead a self-supporting superconductor, $\Delta_r = -U_a \langle c_{r\downarrow} c_{r\uparrow} \rangle$ with U_a chosen such that $|\Delta| = t$, the phase drop spreads over the arms. For the weak link ($t_J = 0.5t$) only 31% of it sits on the junction, and the curvature of the self-consistent $E(\varphi)$ obeys the series rule $1/E''_{sc} = 1/E''_{\text{pinned}} + 1/K$ to 6%, with a stiffness of the 13 arm bonds $K = 0.080t$, i.e. a kinetic inductance $L_K = (\hbar/e)^2/K$. In the loop energy this inductance is in series with that of the mode, and the instability condition becomes $\beta + \beta_K = 1$ with $\beta_K = 4E_J/K = 2\pi L_K I_c / \Phi_0$: 0.57, 0.38, 0.20 and 0.12 at $U = t, 2t, 4t$ and $6t$. In a real device L_K is simply part of L ; the fixed- Δ model isolates the contribution of the mode.

S2. SHORT RING WITH A POLARIZED DOT: BREAKDOWN OF THE PRODUCT STATE

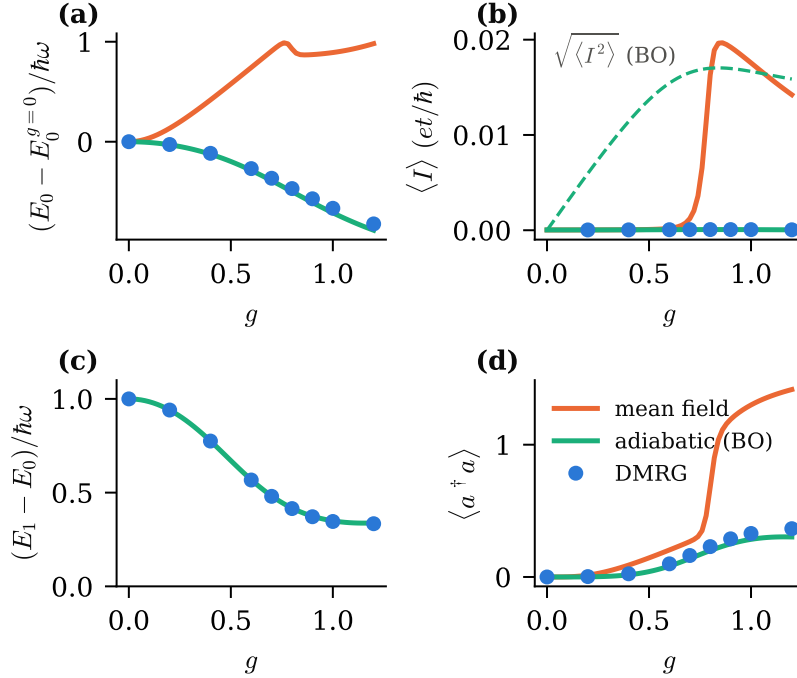


FIG. S1. Short ring with a Zeeman-polarized dot ($t = 2$, $\Delta = 0.6$, $L = 11$, $\epsilon_{d\uparrow} = -0.3$, $\epsilon_{d\downarrow} = 3.7$, $t_J = 0.5$, $\hbar\omega = 0.02$, seed $\varphi_{cl} = 10^{-3}$): mean field (orange), adiabatic theory (green) and DMRG (symbols). (a) Ground-state energy, (b) current (dashed: rms current of the adiabatic ground state), (c) gap to the first excited state, (d) photon number.

Figure S1 shows the model of Ref. [notebook] (Zeeman-polarized dot in a ring of 11 sites, $\hbar\omega = 0.02$). The product state predicts a current-carrying phase above $g \simeq 0.75$ with $\langle a^\dagger a \rangle \simeq 1$ –1.4 and $|\langle I \rangle| \approx 0.02$. DMRG shows that it is an artefact: the exact energy decreases monotonically with g and lies below the MF energy by up to $1.8\hbar\omega$, the gap stays at $0.33\hbar\omega$ ($g = 1.2$), the photon number saturates at 0.36 and the current is the linear response to the seed, 7×10^{-5} . DMRG and the adiabatic theory agree to $0.06\hbar\omega$ in energy and 1.3% in the gap; the residual is the diagonal Born–Oppenheimer correction of order $\hbar\omega/E_A \approx 0.1$. Two effects cause the failure: $E_J \simeq \hbar\omega$ (the Landau double

well appears at $g = 0.58$ but is shallower than the zero-point energy) and the loss of phase-independent kinetic energy in the gauge-non-invariant product state. In addition $L \lesssim \xi$, so $E(\varphi)$ of this ring is dominated by the single-electron harmonic, $E_1 = 2.1 \times 10^{-2} \gg E_2 = 2.0 \times 10^{-3}$ [Fig. 2(a) of the main text]. Reducing $\hbar\omega$ to 0.002 restores a transition in the same model, at $g_c = 0.21$ exactly versus 0.30 in mean field (Fig. S2).

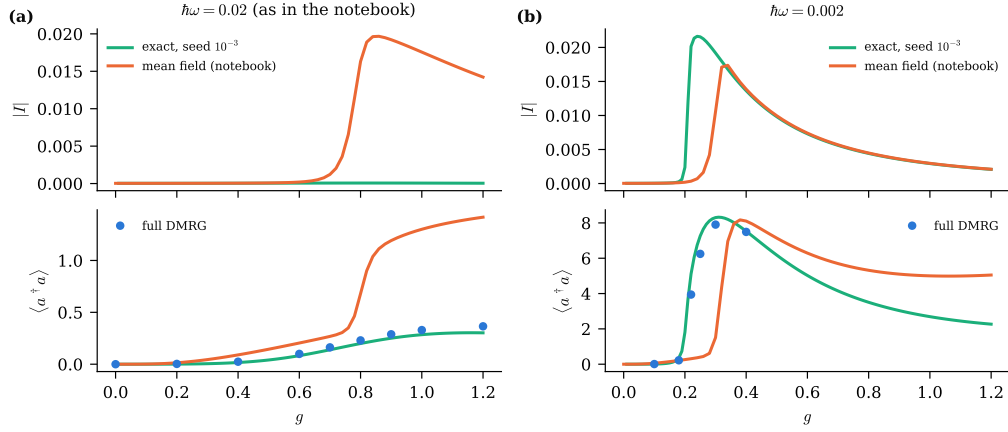


FIG. S2. Same model: current (top) and photon number (bottom) versus g for (a) $\hbar\omega = 0.02$ and (b) $\hbar\omega = 0.002$. Green: exact (adiabatic, seed 10^{-3}); orange: mean field; circles: full DMRG.

S3. HALF-FLUX-BIASED WEAK LINK: A SELECTED RATHER THAN SPONTANEOUS CURRENT

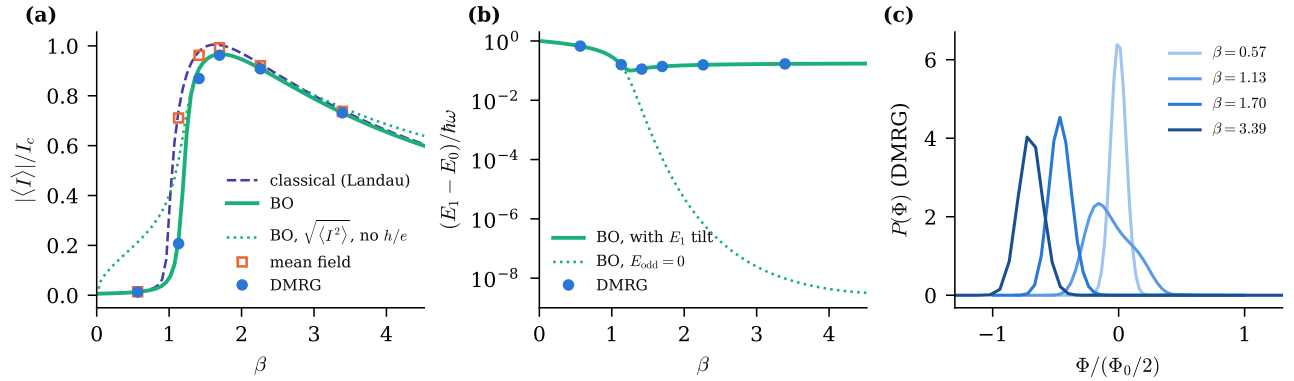


FIG. S3. Weak-link loop ($t_J = 0.5$, $E_2 = -0.056t$, Andreev gap $0.77t$) biased by $\Phi_0/2$, $E_J/\hbar\omega = 10$. (a) Persistent current versus β : DMRG (circles), MF (squares), adiabatic theory with (solid) and without (dotted) the odd harmonics, classical Landau theory (dashed). (b) Splitting of the two lowest levels. (c) Flux distribution of the DMRG ground state.

For an ordinary weak link biased by $\varphi_{cl} = \pi/2$ (Fig. S3) the current rises at $\beta \simeq 1.1$, saturates at $0.97 I_c$ near $\beta = 1.7$ and then decreases as the flux approaches $\Phi_0/2$. DMRG and adiabatic theory agree to $0.003\hbar\omega$ in energy, 5% in current and 2% in $E_1 - E_0$; the product state is $0.2\text{--}0.4\hbar\omega$ too high and overestimates the current threefold near threshold ($0.71 I_c$ versus $0.21 I_c$ at $\beta = 1.13$), becoming accurate only deep in the ordered phase where the photon is nearly coherent. The flux distribution evolves from a Gaussian into a *single* displaced peak: the residual odd harmonic $E_1 = -5.4 \times 10^{-4}t$ tilts the double well by $2|E_1| \sin \varphi^* \approx 0.2\hbar\omega$, which fixes the current direction and sets $E_1 - E_0$ for $\beta > 1.2$ in DMRG and adiabatic theory alike (without it $\delta/\hbar\omega$ would drop below 10^{-6} at $\beta = 2$). In a homogeneous ring this tilt is $\propto e^{-L/\xi}$; in practice it is set by the flux offset.

S4. NUMERICAL DETAILS

Current from DMRG. In the ordered phase the two lowest states of the full problem are degenerate to $\delta \ll$ the DMRG energy resolution, and the variational ground state is an uncontrolled superposition of them. One-site observables that are even under time reversal ($\langle a^\dagger a \rangle$, $\langle X^2 \rangle$) and the doublet quantities (δ , $|\langle 0|X|1 \rangle|$) are therefore robust, whereas $\langle X \rangle$ and $\langle I \rangle$ of a seeded run are reliable only where the seed energy $I^* \varphi_{\text{cl}}$ exceeds both δ and the energy resolution. The three-site bond-current operator is in addition a small difference of large terms and converges poorly at $\chi = 256$; we use Eq. (5) throughout. *Harmonics.* In the doublet sector a six-harmonic cosine fit of $E(\varphi)$ on 9–17 phases reproduces the DMRG energies to $10^{-7}t$. The singlet sector has a cusp at $\varphi = \pi/2$ (level crossing); since this sector is diamagnetic and the photon stays within $|gX| < \pi/2$, it is interpolated on $[0, \pi/2]$ only. Between the 16 DMRG values of U the harmonics are interpolated with monotone cubic splines. *Classical line.* $g_c^{\text{cl}}(U) = [\hbar\omega/2|E''(0; U)|]^{1/2}$ in the doublet sector. *DMRG resolution of δ .* At $U = 2t$, $\beta = 1.5$, DMRG gives $\delta = 6 \times 10^{-6}t$ and $3 \times 10^{-5}t$ at $\chi = 256$ and 400, against $5 \times 10^{-7}t$ from the adiabatic theory; we quote $\sim 3 \times 10^{-5}t$ as the resolution.